

Mobile Vaccination Team Weekly Scheduling and Routing with Long-Term Fairness Considerations

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Abstract

We consider the problem of multi-day scheduling and routing with multi-type, multi-nurse coordination, with depot constraints and with nurse starting and ending each day at their homes. We address the gap in existing vehicle routing problem and workforce scheduling literature that 1) ignores nurses traveling from homes, and 2) ignores multi-period fairness. Our method uses a mixed-integer linear programming formulation for the weekly integrated problem along with both myopic and cumulative fairness modeling. We solve this problem using decomposition methods and heuristics. Our results show that with a small increase in travel cost we can achieve a large fairness improvement in both myopic and cumulative. In summary, our contribution is in developing a model for weekly routing and scheduling for home-based nurses with fairness consideration, and developing an fast algorithm suitable for this problem at a realistic scale. This provides a practical framework for the operation of mobile vaccination teams.

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1 Introduction

Motivation Mobile vaccination / healthcare outreach logistics Workforce satisfaction $\rightarrow$ retention $\rightarrow$ service quality Gap Classical VRP: cost-focused, single-period Workforce scheduling: fairness considered, but not routing-integrated + multi-period Contributions New integrated multi-day routing + assignment model Formalization of long-term fairness Comparison: cost vs fairness tradeoff Computational insights on tractability

2 Literature Review

2.1 Vehicle Routing with Workforce Considerations

The Home Health Care Routing and Scheduling Problem (HHCSP) is a prominent extension of the classical vehicle routing problem that integrates routing decisions with workforce assignment and scheduling. In this setting, nurses or caregivers must be assigned to geographically distributed service locations while respecting time windows, skill requirements, and labor constraints. As a result, HHCSP combines elements of routing, scheduling, and personnel assignment, making it significantly more complex than standard vehicle routing problems [2, 8].

A wide range of objectives have been considered in the HHCSP literature, including minimizing travel cost, improving service quality, and ensuring workforce feasibility. In particular, many studies incorporate workforce-related considerations such as skill matching, synchronization of multiple caregivers, and workload balancing [7, 4]. Multi-objective formulations are commonly adopted, where travel efficiency is optimized jointly with workforce-related criteria [5].

Despite these advances, most HHCSP models are inherently patient-centric, often incorporating continuity-of-care requirements that favor assigning the same caregiver to the same patient across multiple visits. In contrast, our problem focuses on assigning nurses to community vaccination events, where such continuity requirements are absent. Instead, the problem emphasizes flexible temporal assignment within a day and coordination of multiple nurses per event, which differentiates it from traditional HHCSP formulations.

2.2 Multi-Period Routing and Scheduling

Multi-period extensions of HHCSP consider planning decisions across multiple days or weeks. These problems are typically addressed using either integrated multi-period formulations or decomposition approaches that separate weekly assignment from daily routing decisions [12]. Such models capture temporal dependencies and allow for more realistic workforce planning.

However, the focus of multi-period routing and scheduling has largely been on operational efficiency and feasibility, such as ensuring coverage and minimizing costs over time. When temporal consistency is considered, it is typically motivated by patient-related objectives, such as maintaining nurse-patient relationships [6].

In contrast, the literature has paid limited attention to worker-centric fairness across periods. In particular, there is little work on how responsibilities or burdens should be distributed among workers over time. This gap is especially relevant in settings such as vaccination operations, where repeated assignments can lead to inequities in workload or responsibilities if not properly managed.

2.3 Fairness in Workforce Scheduling and Routing

Fairness has been extensively studied in the nurse scheduling (rostering) literature, where it is recognized as a key factor affecting staff satisfaction, retention, and well-being [11, 3]. The most

common notion of fairness in this context is workload balance, which aims to distribute working hours, shifts, or tasks evenly across nurses. This is typically modeled using objectives such as minimizing the difference between maximum and minimum workload, minimizing variance, or enforcing bounds on individual workloads [9].

In addition to workload balance, fairness in nurse scheduling often involves the equitable distribution of undesirable assignments, such as night shifts or weekend duties, as well as the consideration of individual preferences [10]. These aspects are usually incorporated as soft constraints or secondary objectives within multi-objective optimization frameworks [1].

In contrast, fairness considerations in routing problems are more limited. Existing studies primarily focus on balancing workload or travel time among workers, often by minimizing the maximum workload or equalizing route lengths [5]. While such approaches capture a notion of equity within a single planning horizon, they do not address how fairness should evolve over time.

2.4 Research Gap and Contribution

The existing literature highlights several important gaps. First, fairness in routing and scheduling is predominantly defined in terms of workload balance within a single planning period. Second, multi-period models rarely incorporate worker-centric fairness, focusing instead on operational efficiency or patient-related objectives. Third, there is little attention to role-based fairness, where different types of responsibilities (e.g., leadership or coordination roles) may impose unequal burdens on workers.

This paper addresses these gaps by introducing two complementary notions of fairness in a weekly nurse routing and scheduling problem for community vaccination operations. The first is within-week workload fairness, which ensures a balanced distribution of working hours across nurses. The second is a novel notion of cumulative leadership fairness, which tracks and balances leadership assignments over time. Unlike traditional workload-based fairness, leadership fairness captures differences in responsibility rather than task quantity, and explicitly accounts for fairness accumulation across multiple planning periods.

By integrating routing, scheduling, and dynamic fairness considerations into a unified deterministic mixed-integer programming framework, this work contributes to the emerging literature on workforce-centric fairness in operational decision-making.

3 Problem Description

There are two types of nurses: Registered Nurses (RNs) and Licensed Vocational Nurses (LVNs). Each event has a fixed date, known location, known duration, and associated required number of RNs and LVNs. For each event, one leader will be selected to pick up and drop off the vaccines and equipments at the depot, as the first and the last thing in the day. The planning horizon is one week, where we need to assign event to time slots, assign nurses to events, and pick the leader for each event. The objective is to minimize the travel cost and ensure fairness across nurses over time.

4 Problem Formulation

4.1 Base Model M0

Model Overview We consider a multi-day scheduling and routing problem for mobile vaccination teams. A set of nurses must be assigned to events over a planning horizon of multiple days.

Each event requires a specified number of nurses with potentially different skill types (RN or LVN), and each nurse starts and ends their route at their home location. The objective is to minimize total travel time while satisfying operational constraints.

Sets

- $\mathcal{I}$: set of events
- $\mathcal{H}$: set of homes
- $\mathcal{L}$: set of locations, including events, homes, and depots
- $\mathcal{W}$: set of nurses. $\mathcal{W}^{RN}$ denotes the set of RNs, and $\mathcal{W}^{LVN}$ denotes the set of LVNs
- $\mathcal{D}$: set of days

Parameters

- c_{ij} : travel time between locations i and j
- c_i : duration of event i
- N_i^{RN}, N_i^{LVN} : minimum number of RNs and LVNs needed for event i
- l_{id}, u_{id} : the earliest and latest feasible starting time for event i on day d

Decision Variables

- $x_{ijd}^w \in \{0, 1\}$: equals 1 if nurse w travels from i to j on day d
- $s_{id} \in \{0, 1\}$: equals 1 if event i is scheduled on day d
- t_{id} : start time of event i on day d
- α_{id}^w : equals 1 if nurse w is assigned to be the leader for event i on day d

Objective Function

$$\min \sum_{i,j \in \mathcal{L}} c_{i,j} \sum_{w \in \mathcal{W}, d \in \mathcal{D}} x_{ijd}^w \quad (1)$$

The objective minimizes the total travel time incurred by all nurses.

Constraints

Event Assignment Constraints

$$\sum_{d \in \mathcal{D}} s_{id} = 1 \quad \forall i \quad (2)$$

Each event must be assigned according to the planning requirements.

Routing Flow Constraints

$$\sum_{j \in \mathcal{L}} x_{ijd}^w \leq s_{id} \quad \forall w \in \mathcal{W}, i \in \mathcal{I}, d \in \mathcal{D} \quad (3)$$

$$\sum_{i \in \mathcal{L}} x_{ijd}^w = \sum_{k \in \mathcal{L}} x_{jkd}^w \quad \forall w \in \mathcal{W}, j \in \mathcal{I}, d \in \mathcal{D} \quad (4)$$

$$\sum_{j \in \mathcal{L}} x_{ijd}^w \leq 1 \quad \forall i \in \mathcal{H} \quad (5)$$

These constraints ensure flow conservation for each nurse.

Time Feasibility Constraints

$$l_{id} \cdot s_{id} \leq t_{id} \leq u_{id} \cdot s_{id} \quad \forall i \in \mathcal{I}, d \in \mathcal{D} \quad (6)$$

$$t_{jd} \geq t_{id} + c_i + c_{ij} - M(1 - x_{ijd}^w) \quad \forall i, j \in \mathcal{I}, d \in \mathcal{D}, w \in \mathcal{W} \quad (7)$$

These constraints enforce temporal consistency along each route. M denotes a number sufficiently large as in the big-M method.

Staffing Constraints

$$\sum_{w \in \mathcal{W}^{RN}, i \in \mathcal{L}, d \in \mathcal{D}} x_{ijd}^w \geq N_j^{RN} \quad \forall j \in \mathcal{I} \quad (8)$$

$$\sum_{w \in \mathcal{W}^{LVN}, i \in \mathcal{L}, d \in \mathcal{D}} x_{ijd}^w \geq N_j^{LVN} \quad \forall j \in \mathcal{I} \quad (9)$$

Each event must be staffed with the required number and type of nurses.

Depot Constraints

$$\sum_{w \in \mathcal{W}, d \in \mathcal{D}} \alpha_{id}^w = 1 \quad \forall i \in \mathcal{I} \quad (10)$$

$$\alpha_{id}^w \leq \sum_{j \in \mathcal{L}} x_{ijd}^w \quad i \in \mathcal{I}, w \in \mathcal{W}, d \in \mathcal{D} \quad (11)$$

$$\alpha_{id}^w \leq x_{HDd}^w \quad \forall i \in \mathcal{I}, w \in \mathcal{W} \quad (12)$$

$$(13)$$

These constraints capture depot-related requirements and other operational rules.

Discussion The base model captures the integrated scheduling and routing decisions for mobile vaccination teams over multiple days. It serves as the benchmark model focusing solely on operational efficiency, without incorporating fairness considerations. In the following sections, we extend this formulation to account for fairness across nurses over time.

4.2 Fairness Model

To account for fairness in work time, we try to avoid schedules that make some nurse work too much and some too little.

Auxiliary Quantities The total travel cost is given by:

$$C^{\text{travel}} = \sum_{i,j \in \mathcal{L}} c_{ij} \left(\sum_{w \in \mathcal{W}, d \in \mathcal{D}} x_{ijd}^w \right) \quad (14)$$

Let the total working hours assigned to nurse w be:

$$H_w = (\text{total working time of nurse } w \text{ over the planning horizon}) \quad (15)$$

Let the total number of leadership assignments for nurse w be:

$$L_w = (\text{total number of leader roles assigned to nurse } w) \quad (16)$$

In the following models, we modify the objective function and add additional constraints for the auxiliary quantities.

Model M1 (Workload Fairness)

$$\min \quad C^{\text{travel}} + \lambda (H_{\max} - H_{\min}) \quad (17)$$

$$\text{s.t.} \quad H_{\max} \geq H_w \quad \forall w \in \mathcal{W}, \quad (18)$$

$$H_{\min} \leq H_w \quad \forall w \in \mathcal{W}. \quad (19)$$

4.2.1 Long-term Fairness Model

In the long-term model, we consider not only the workload and leadership in the current week, but also for previous weeks. We construct a cumulative fairness penalty term to penalize repeated assignment of leaders to the same person over the weeks. Fairness in a longer time span is important because it allows for some imbalance within the week but aims to ensure fairness in the long run.

Model M2 (Long-Term Fairness with Leadership) For each week T , solve the problem

$$\min \quad C^{\text{travel}} + \lambda (H_{\max}^T - H_{\min}^T) + \mu (L_{\max}^T - L_{\min}^T) \quad (20)$$

where

- λ, μ are scaling parameters
- $L_w^T = \eta_1 \cdot (\# \text{ events led by } w \text{ in week } T) + \eta_2 \cdot (\# \text{ leader-days of } w \text{ in week } T)$, and let L_w^T denote the sum of L_w^t up to and including week T .
- H_w^T is the total working time assigned to nurse w in week T ,
- $H_{\max}^T = \max_w H_w^T$ and $H_{\min}^T = \min_w H_w^T$,
- $L_{\max}^T = \max_w L_w^T$ and $L_{\min}^T = \min_w L_w^T$

Work hours are naturally evaluated over short operational cycles because staff experience workload week by week; leadership assignments are less frequent, more discrete, and path-dependent, so cumulative imbalance is the more relevant fairness notion.

4.3 Discussion

modeling choices: fairness metric (variance / max-min / Gini-like), trade-off parameter. Plot pareto front with varying λ .

5 Solution Approach

5.1 Solving the Base Model

To solve the base model at a realistic scale and under time constraints, we introduce the following additional constraint.

Events start at full or half hour. Under a realistic operational setting, vaccination events only starts at half-hour or full-hours. Hence, we add these constraints.

$$t_{id} = 30 \cdot b_{id}, \quad b_{id} \in \mathbb{N} \quad \forall i \in \mathcal{I}, d \in \mathcal{D} \quad (21)$$

Events has one feasible day. In the real operational setting, the venue requesting a vaccination event will pick the one day that they prefer the most. This essentially fixes s_{id} and makes the weekly problem decomposable by day.

5.2 Solving the Fairness Model

Note that the problem is decomposable by day except $H_{\max}, H_{\min}, L_w$.

6 Computational Experiments

7 Fairness Experiments

The goal of these experiments is to compare model performance under different fairness criteria. Our baseline model ($M0$) considers no explicit fairness terms and simply minimizes the total travel cost.

The second model ($M1$) incorporates *myopic fairness*. In addition to minimizing travel cost, it includes penalty terms that encourage fairness within a single week.

The third model ($M2$) considers *cumulative fairness*. Instead of evaluating fairness only within the current week, it also accounts for fairness accumulated over previous weeks.

Conceptually, the goal is to test the following hypothesis: if we allow a small increase in total travel cost and evaluate fairness over a longer time horizon, the resulting schedules are significantly more balanced across nurses. In particular, we want to see whether the cumulative fairness approach outperforms the myopic model in the long run.

7.1 Experimental Design

We use the Solomon dataset as the base instance, with evenly distributed time windows (10 events per day). Nurse home locations and the depot location are fixed.

For each simulated week, event locations are randomized by adding noise to the original Solomon coordinates. This simulates the realistic situation where events tend to cluster around certain geographic centers but vary from week to week.

Event time windows, service durations, and nurse requirements will also be randomized within reasonable ranges.

In later experiments, we will also test different spatial distributions of events (e.g., more clustered vs. more dispersed instances).

7.2 Fairness Objectives

Based on feedback from nurses and our understanding of the operational setting, we may consider the following fairness metrics:

- **Total working hours** (more hours correspond to more pay)
- **Total travel time**
- **Number of times assigned as leader** (nurses generally prefer not to be the leader)

The cumulative model ($M2$) incorporates fairness measures accumulated across previous weeks, allowing the model to compensate for past imbalances when generating future schedules.

Specifically, the objective functions are as follows:

- ($M0$) total travel cost
- ($M1$) total travel cost + $\lambda * (\text{max hours assigned} - \text{min hours assigned})$
- ($M2$) total travel cost + $\lambda_w * \text{Penalty}_w * \text{Hours}_w$
- ($M3$) total travel cost + $\lambda_w * \text{Penalty}_w * \text{Hours}_w + \mu_w \cdot \text{leader count } w$

Events	Nurses	Runtime (s)
20	20	12.3
50	50	84.7

Table 1: Runtime comparison

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